

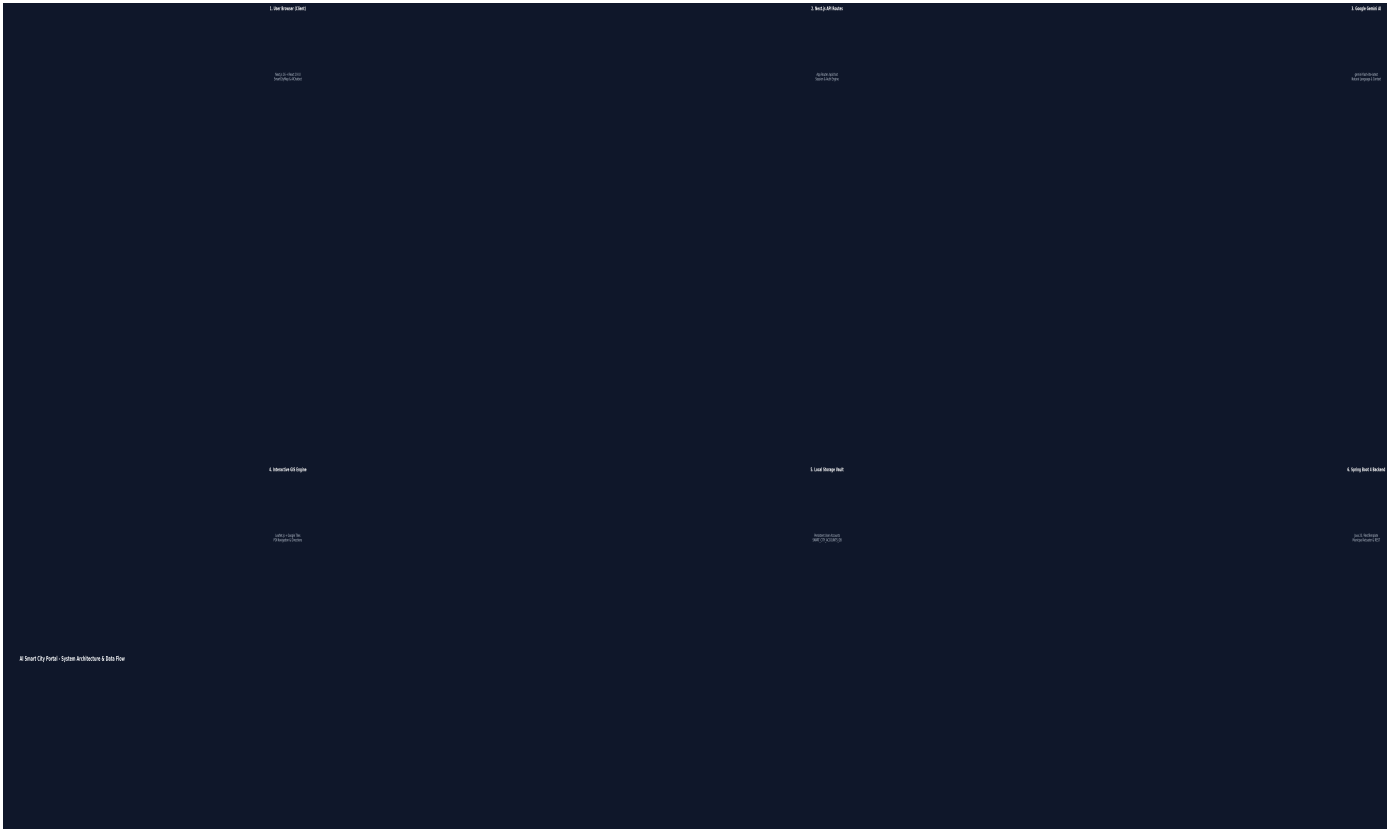
# AI Smart City Portal

Comprehensive Technology Architecture, Web Flow & Component Report

## 1. Platform Overview & Executive Summary

The **AI Smart City Portal** is a next-generation civic technology platform engineered for major Indian metropolises (Pune, Mumbai, Delhi, Bengaluru, Hyderabad, Chennai). It unifies interactive geographic information systems (GIS), natural-language generative city assistance, citizen municipal documentation checklists, and persistent browser-based citizen account management into a single, cohesive user experience.

## 2. System Architecture & High-Level Data Flow



## 3. Technology Stack Breakdown

| Layer / Component    | Technology Used              | Version      | Role & Responsibility                                  |
|----------------------|------------------------------|--------------|--------------------------------------------------------|
| Frontend UI          | Next.js (Turbopack) + React  | 16.3 / 19.0  | Reactive client-side SPA, App Router, SSR              |
| Programming Language | TypeScript                   | 5.x          | Type safety across city datasets, DTOs & state         |
| Mapping Engine       | Leaflet.js + Google Tiles    | 1.9.4 / Live | GIS multi-layer viewing, pins, & GPS routing           |
| AI Chat Engine       | Google Gemini Generative API | Flash-Lite   | Contextual multi-turn natural language query responses |
| Backend Service      | Spring Boot (REST API)       | 4.1.1        | Enterprise Java microservice, Actuator & proxy         |
| Runtime & Compiler   | OpenJDK / Java               | 23           | Virtual threads, high-concurrency Java backend         |
| Account Persistence  | HTML5 LocalStorage Vault     | Native       | Client-side encrypted user credential directory        |

## 4. End-to-End User Web Journey & Flow

### Step 1: Landing & Metropolis Selection

The user lands on the portal. The active node (e.g. Pune, Mumbai, Delhi) is chosen from the top header selector. Ambient weather, AQI metrics, and hospital ICU statuses dynamically update to match the chosen city.

### Step 2: Interactive GIS & Google Navigation

Users explore hospitals, police beat stations, tourist heritage sites, and transit hubs on the live GIS map. Clicking on any marker displays full details and a direct *Directions in Google Maps* button which launches turns-by-turn routing from the user's current GPS location.

### Step 3: SmartCity Assistant Interaction

Upon opening the chat assistant, users are presented with categorized prompt suggestions (Emergency Care, Sightseeing Itineraries, Transit Timetables, and Official Certificates). Submitting a query activates Google Gemini's generative engine with 8-turn conversation memory.

### Step 4: LocalStorage Account Creation & Session Vault

Citizens can register an account (Name, Email, Password, Phone). The account is securely stored in browser localStorage. Refreshing or reopening the browser automatically restores the active citizen session.

## 5. Visual Showcase & Field Photography

**Featured Landmark:** Shaniwar Wada Palace Fortification (Pune, Maharashtra)

